

2018 HKDSE Physics

Paper 2 — Elective Sections

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Section A: Astronomy and Space Science

Multiple-choice questions

1.1

An unpowered spacecraft is travelling in an elliptical orbit around the Earth. Which of the following descriptions is/are correct?

- (1) The acceleration of the spacecraft is always pointing towards the centre of the elliptical orbit.
- (2) The magnitude of the acceleration of the spacecraft is greatest when it is moving fastest.
- (3) The gravitational potential energy of the spacecraft remains unchanged in the orbit.

- A (1) only
- B (2) only**
- C (1) and (3) only
- D (2) and (3) only

Official answer: B

The acceleration is directed towards the Earth at a focus, not the centre of the ellipse. Its magnitude GM/r^2 is greatest at periapsis, where the spacecraft is fastest, while $U = -GMm/r$ changes with r . Only (2) is correct.

Answer: B.

1.2

The escape velocity at the Earth's surface is 11.2 km s^{-1} . A space probe launched from the Earth's surface attains a speed of 6 km s^{-1} when it is very far away from the Earth. Find the launching speed of the space probe. Assume that the flight is unpowered except during launching and neglect the effects of other celestial bodies.

- A 12.7 km s^{-1}
- B 15.6 km s^{-1}
- C 16.4 km s^{-1}
- D 17.2 km s^{-1}

Official answer: A

Energy conservation gives $u^2 = v_{\text{escape}}^2 + v^2 = 11.2^2 + 6.0^2$. Hence $u = 12.7 \text{ km s}^{-1}$. Answer: A.

1.3

Which of the following observations can be explained by both Ptolemy's geocentric model and Copernicus' heliocentric model?

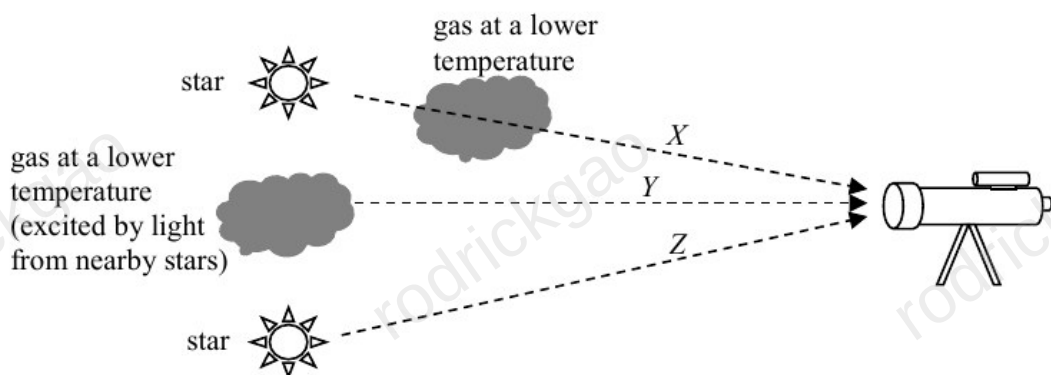
- (1) The brightness of planets varies throughout the year.
 - (2) The retrograde motion of Mars in the sky.
 - (3) Venus shows a complete cycle of phases in the sky.
- A (1) only
 - B (3) only
 - C (1) and (2) only
 - D (2) and (3) only

Official answer: C

Both models can reproduce changes in planetary brightness and apparent retrograde motion; Ptolemy's model cannot reproduce Venus's complete cycle of phases. Thus (1) and (2) are common to both. Answer: C.

1.4

A telescope obtains spectra from light coming from three directions X, Y and Z. Which of the following gives the spectra obtained?



	Continuous spectrum	Absorption-line spectrum	Emission-line spectrum
A	direction Z	direction X	direction Y
B	direction Z	direction Y	direction X
C	direction X	direction Y	direction Z
D	direction X	direction Z	direction Y

Official answer: A

A hot dense source gives a continuous spectrum; viewing it through cooler gas gives absorption lines, while excited low-density gas gives emission lines. These correspond respectively to Z, X and Y. Answer: A.

1.5

A star of radius R has a parallax of θ when observed from the Earth. Which of the following gives the angular size of the star with respect to an observer on the Earth?

- A $4R\theta / 1 \text{ AU}$
- B $2R\theta / 1 \text{ AU}$
- C $R\theta / 1 \text{ AU}$
- D $R\theta / 2 \text{ AU}$

Official answer: B

For small angles, parallax $\theta = 1 \text{ AU}/d$ and angular diameter $\alpha = 2R/d$. Therefore $\alpha = 2R\theta/(1 \text{ AU})$. Answer: B.

1.6

The table below shows the apparent magnitudes and absolute magnitudes of four stars. Which star is farthest from the Earth?

Star	Apparent magnitude	Absolute magnitude
P	0	−0.5
Q	−0.5	2
R	2	−1.5
S	−2	−2.5

A P

B Q

C R

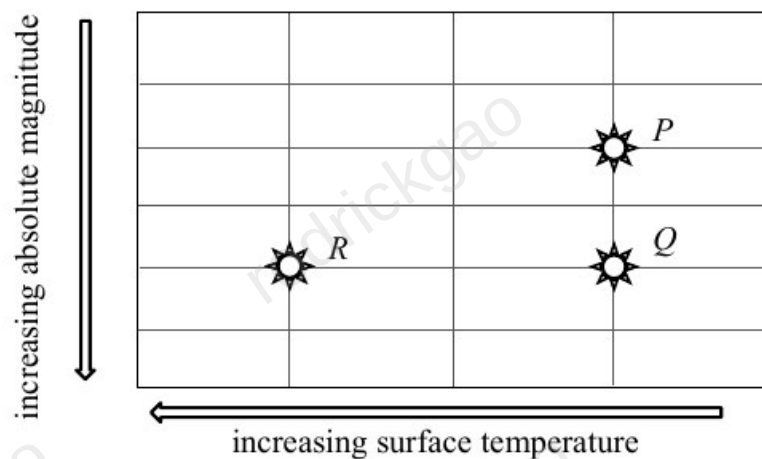
D S

Official answer: C

Distance modulus increases with distance: $m - M = 5 \log_{10}(d/10 \text{ pc})$. The values are 0.5, −2.5, 3.5 and 0.5 for P, Q, R and S, so R is farthest. Answer: C.

1.7

The information about the absolute magnitude and surface temperature of stars P, Q and R are shown below. If Q belongs to the main sequence, arrange P, Q and R in ascending order of their radii.



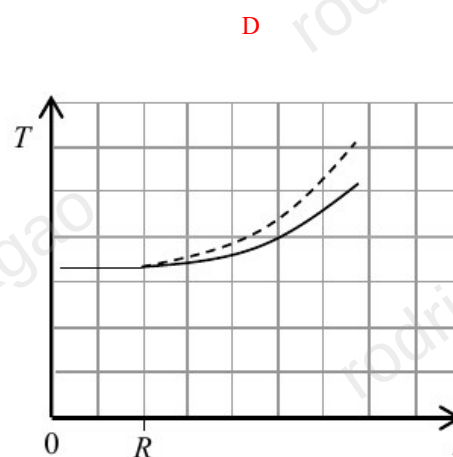
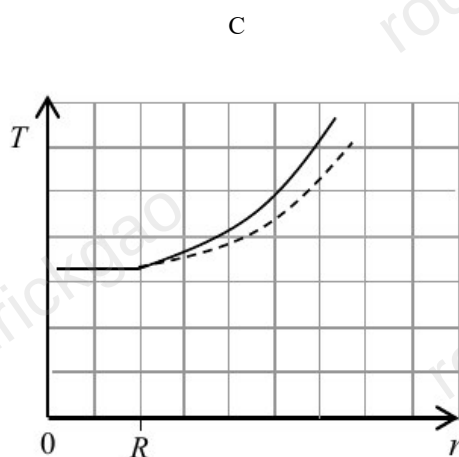
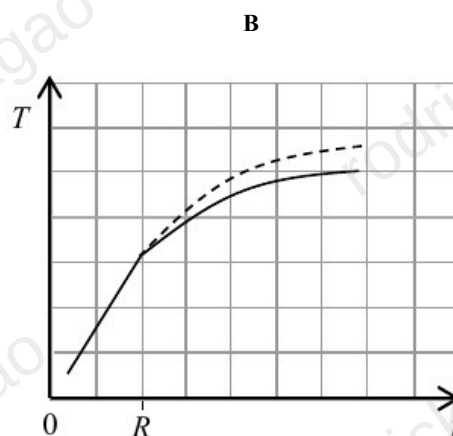
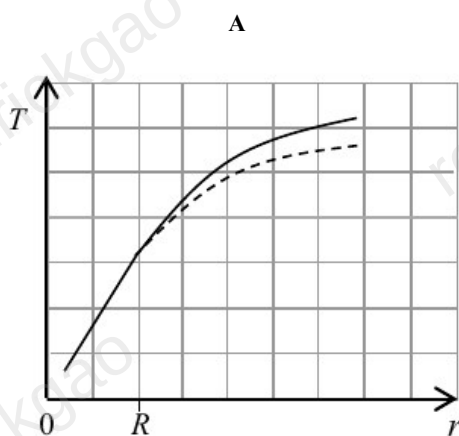
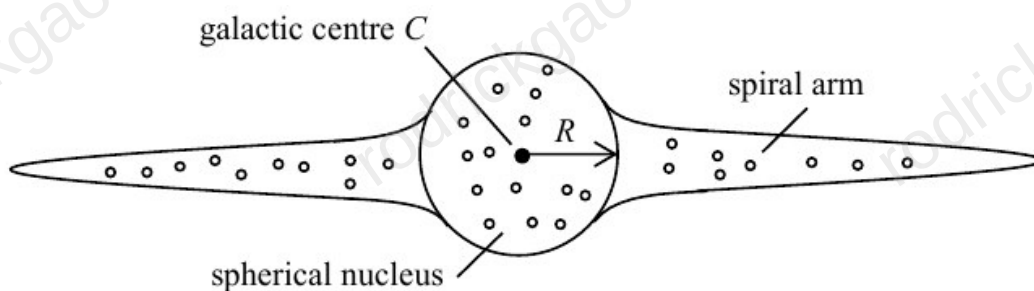
- A $Q < R < P$
- B $R < P < Q$
- C $P < R < Q$
- D $R < Q < P$

Official answer: D

Using $L \propto R^2 T^4$, or $R \propto \sqrt{L/T^2}$, with the plotted temperatures and absolute magnitudes gives the ascending radius order $R < Q < P$. Answer: D.

1.8

The figure shows the side view of a typical galaxy with a spherical nucleus of radius R and several spiral arms. Observations show that the stars in the nucleus are rotating about the galactic centre C as if the nucleus is a solid body. It is expected that the rotation of the stars in the spiral arms about C will obey Kepler's third law but instead the stars are observed to rotate with nearly the same constant speed. Which graph best shows the variation of the rotational period T of the stars with their distance r from C ? (The solid curve represents the observed result whereas the dotted curve represents the expected result.)



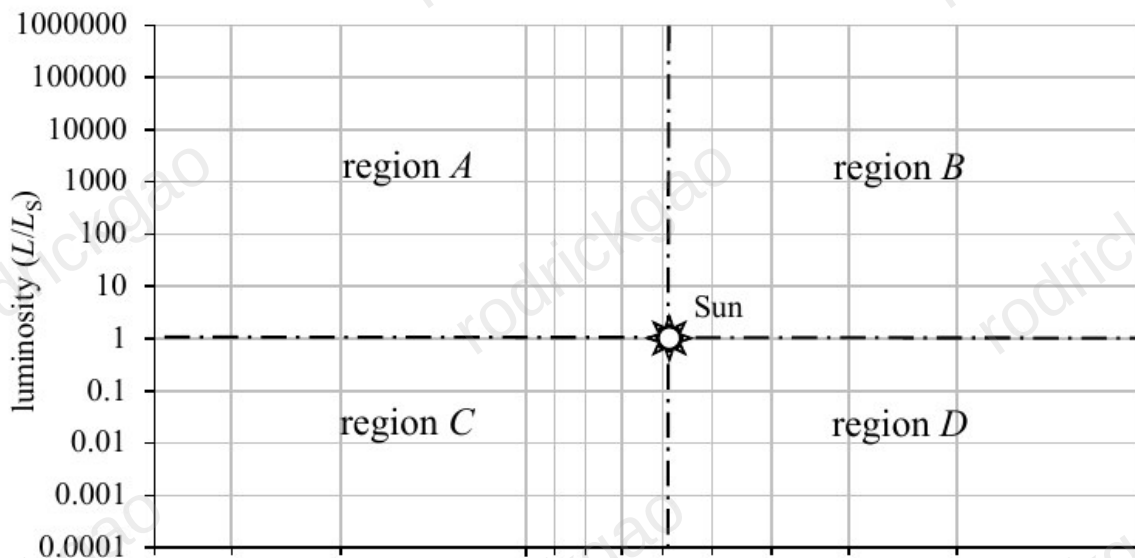
Official answer: D

Solid-body rotation in the nucleus gives constant angular speed and hence constant T . Outside, the observed nearly constant speed gives $T \propto r$, whereas Kepler's law predicts $T \propto r^{3/2}$; graph D shows these trends.

Answer: D.

Question 1: Structured question

- (a) X is a distant star and it has a parallax of $0.08''$ when observed from the Earth.
- (i) Find the distance of star X from the Earth in ly. (2 marks)
- (ii) The information of star X is given below: surface temperature = 2900 K; radius = $0.14 R_s$, where R_s is the radius of the Sun. Given the surface temperature of the Sun $T_s = 5800$ K, deduce the luminosity of star X in terms of the luminosity of the Sun, L_s . Hence, or otherwise, state in which region, A, B, C or D, on the Hertzsprung–Russell (H–R) diagram X is located. (3 marks)



Hertzsprung–Russell diagram

- (b) Star X has an Earth-sized planet Y orbiting around it.
- (i) Why is it difficult to observe Y directly using optical telescopes other than it is very small and extremely far away from the Earth? (1 mark)
- (ii) Astronomers are able to deduce the orbital period T of Y around its parent star X indirectly since the interaction between them causes X to wobble (i.e. small degree of rotation about their centre of mass) with the same period. Name the kind of interaction and state what physical quantity of X should be measured in order to find T . (2 marks)
- (iii) It has been postulated that a favourable condition for life to exist on a certain planet is that the power per unit area coming from the parent star onto that planet I_Y is within 0.5 to 2 times that coming from the Sun onto the Earth I_E . Deduce whether this condition is satisfied for planet Y using the result in (a)(ii). Given that the distance between X and Y is 0.04 AU. (Note: $I = L/(4\pi d^2)$, where L is the luminosity of the parent star and d is the distance of the planet from its parent star.) (2 marks)

Official marking scheme

Solution		Marks	Remarks
1. (a)	(i) Distance from the Earth $= \frac{1}{0.08} \text{ pc} (=12.5 \text{ pc} = 12.5 \times 3.26 \text{ ly})$ $= 40.75 \text{ (ly)}$	1M 1A	
		2	
	(ii) $L = 4\pi R^2 \sigma T^4$ $\therefore L \propto R^2 T^4$ OR $\frac{L}{L_s} = \left(\frac{R}{R_s}\right)^2 \left(\frac{T}{T_s}\right)^4$ $\frac{L}{L_s} = \left(\frac{0.14}{1}\right)^2 \left(\frac{2900}{5800}\right)^4$ $= 0.001225$ ≈ 0.00123 Region D	1M 1A 1A	
		3	
	(b) (i) Planet does not emit light / only reflects light, so easily overshadowed by the parent star's flare.	1A	
		1	
	(ii) Gravitation / gravitational (force of) attraction / gravity. By monitoring the Doppler shift of a particular spectral line (from X) caused by the variation of the star's radial velocity, period T can be found.	1A 1A	
		2	
	(iii) $I = \frac{L}{4\pi d^2}$ $\therefore I \propto \frac{L}{d^2}$ OR $\frac{I_Y}{I_E} = \left(\frac{L_X}{L_s}\right) \left(\frac{d_E}{d_Y}\right)^2$ $\frac{I_Y}{I_E} = \left(\frac{0.001225}{1}\right) \left(\frac{1}{0.04}\right)^2$ $= 0.765625$ $\approx 0.766 \text{ (within 0.5 to 2)}$ The planet satisfied the condition for the existence of life.	1M 1A	
		2	

(a) (i) $d = 1/p = 1/0.08 = 12.5 \text{ pc} = 12.5 \times 3.26 = 40.8 \text{ ly}$.

(ii) $L_X/L_s = (0.14)^2 (2900/5800)^4 = 0.001225$, placing X in region D.

(b) (i) The planet emits negligible light and its reflected light is overwhelmed by the star.

(ii) The interaction is gravitational; monitor the periodic Doppler shift/radial velocity of X.

(iii) $I_Y/I_E = 0.001225/(0.04)^2 = 0.766$, which lies between 0.5 and 2, so the stated condition is satisfied.

Section B: Atomic World

Multiple-choice questions

2.1

In Rutherford's scattering experiment, the distance of closest approach of an α particle to a gold atom can be deduced. This distance is a good estimate of the upper limit of

- A the radius of an α particle.
- B the radius of a gold atom.
- C the radius of a gold nucleus.
- D the thickness of the gold foil.

Official answer: C

The closest approach is the point where the α particle's kinetic energy has been converted into electric potential energy without entering the nucleus. It therefore gives an upper limit to the gold-nucleus radius. Answer: C.

2.2

Which of the following is/are the assumption(s) of the Bohr model of an atom?

- (1) The orbital radii of the electrons are quantized.
- (2) The electric force between the nucleus and an electron provides the centripetal force for the circular motion.
- (3) The total energy of an orbiting electron remains unchanged.

- A (2) only
- B (3) only
- C (1) and (2) only
- D (1), (2) and (3)

Official answer: D

In the Bohr model, allowed orbits are quantized; electrostatic attraction supplies centripetal force; and an electron in a stationary orbit has constant total energy. All three statements are assumptions. Answer: D.

2.3

Which statements below about spectra are correct?

- (1) A steel rod heated to white hot emits a continuous spectrum.
- (2) The dark lines in the absorption spectrum of an element match in frequency with the bright lines in this element's emission spectrum.
- (3) The atomic spectrum of an element is a piece of evidence for the existence of energy levels in its atoms.

- A (1) and (2) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: D

A hot solid produces a continuous spectrum; an element's absorption and emission lines occur at the same transition frequencies; and discrete spectra evidence atomic energy levels. All three are correct. Answer: D.

2.4

According to the Bohr model of an atom, when the electron of a hydrogen atom undergoes transition from an inner orbit to an orbit of larger radius, the hydrogen atom may have

- A absorbed a photon, and the electron's kinetic energy decreases.
- B absorbed a photon, and the electron's kinetic energy increases.
- C emitted a photon, and the electron's kinetic energy decreases.
- D emitted a photon, and the electron's kinetic energy increases.

Official answer: A

A transition to a larger orbit requires photon absorption. For a Bohr orbit, $K \propto 1/n^2$, so the electron's kinetic energy decreases as n increases. Answer: A.

2.5

Some hydrogen atoms in the second excited state (i.e. $n = 3$) subsequently produce a spectrum consisting of a series of discrete lines. How many spectral lines of different wavelengths are there in the spectrum?

- A 1
- B 2
- C 3
- D 4

Official answer: C

From $n = 3$ the possible downward transitions are $3 \rightarrow 2$, $3 \rightarrow 1$ and $2 \rightarrow 1$, giving three distinct photon energies and wavelengths. Answer: C.

2.6

Electrons, each of mass m and charge e , are being accelerated in a transmission electron microscope (TEM). Estimate the accelerating voltage of the TEM if the de Broglie wavelength of the electrons is λ .

- A $h^2/(m\epsilon\lambda^2)$
- B $h^2/(2m\epsilon\lambda^2)$
- C $m\epsilon\lambda^2/h^2$
- D $2m\epsilon\lambda^2/h^2$

Official answer: B

$eV = p^2/(2m)$ and $p = h/\lambda$. Therefore $V = h^2/(2m\epsilon\lambda^2)$. Answer: B.

2.7

Which of the following statements about a scanning tunnelling microscope (STM) is/are correct?

- (1) It works like an optical microscope except that a high energy electron beam is used instead of visible light and a magnetic field acts as a lens.
- (2) It can be used to show the arrangement of atoms at a conductive surface.
- (3) The magnitude of the tunnelling current depends on the separation between the surface under investigation and the probe of the STM.

- A (1) only
- B (3) only
- C (2) and (3) only**
- D (1), (2) and (3)

Official answer: C

Statement (1) describes an electron microscope, not an STM. An STM maps atoms on a conducting surface, and its tunnelling current depends very strongly on probe–surface separation; (2) and (3) are correct. Answer: C.

2.8

Which statement below is INCORRECT?

- A Nanoparticles of silver show a colour different from the colour of silver in bulk form.
- B Diamond is a poor thermal conductor and a poor electrical conductor as it does not have free electrons.**
- C A carbon nanotube is much stronger than steel of the same size.
- D Carbon buckyballs such as C_{60} are formed by carbon atoms arranged in hollow cages.

Official answer: B

Diamond is a good thermal conductor even though it is a poor electrical conductor. The other statements correctly describe nanosilver, nanotubes and C_{60} , so B is the incorrect statement. Answer: B.

Question 2: Structured question

The set-up shown in Figure 2.1 can be used to measure the maximum kinetic energy of photoelectrons emitted from sodium metal when electromagnetic radiations of the same intensity but having different frequencies are incident on a sodium-coated cathode.

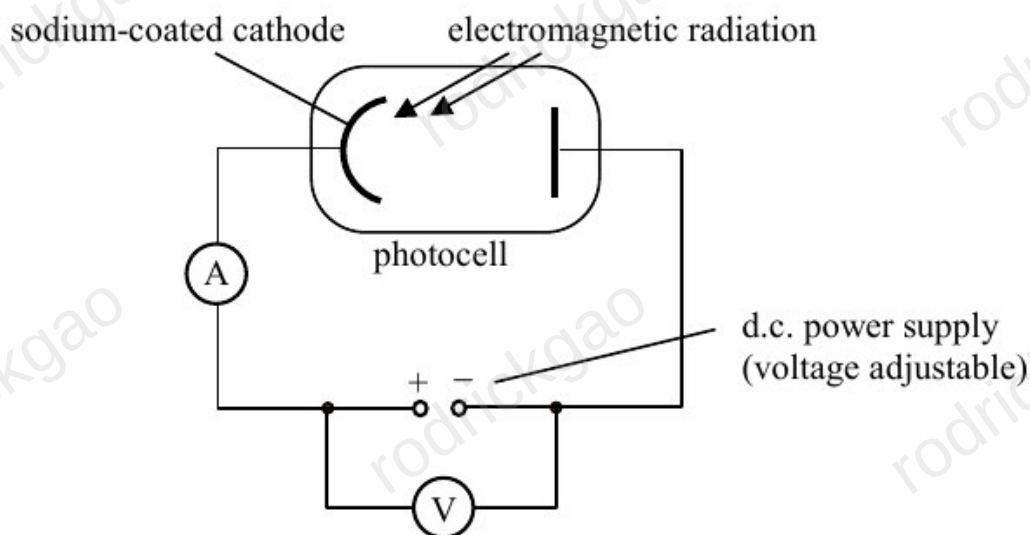


Figure 2.1

- (a) Briefly describe how the maximum kinetic energy of photoelectrons can be measured using this set-up for a certain frequency (above the threshold value) of radiation. (2 marks)
- (b) The graph in Figure 2.2 shows the results of the experiment.

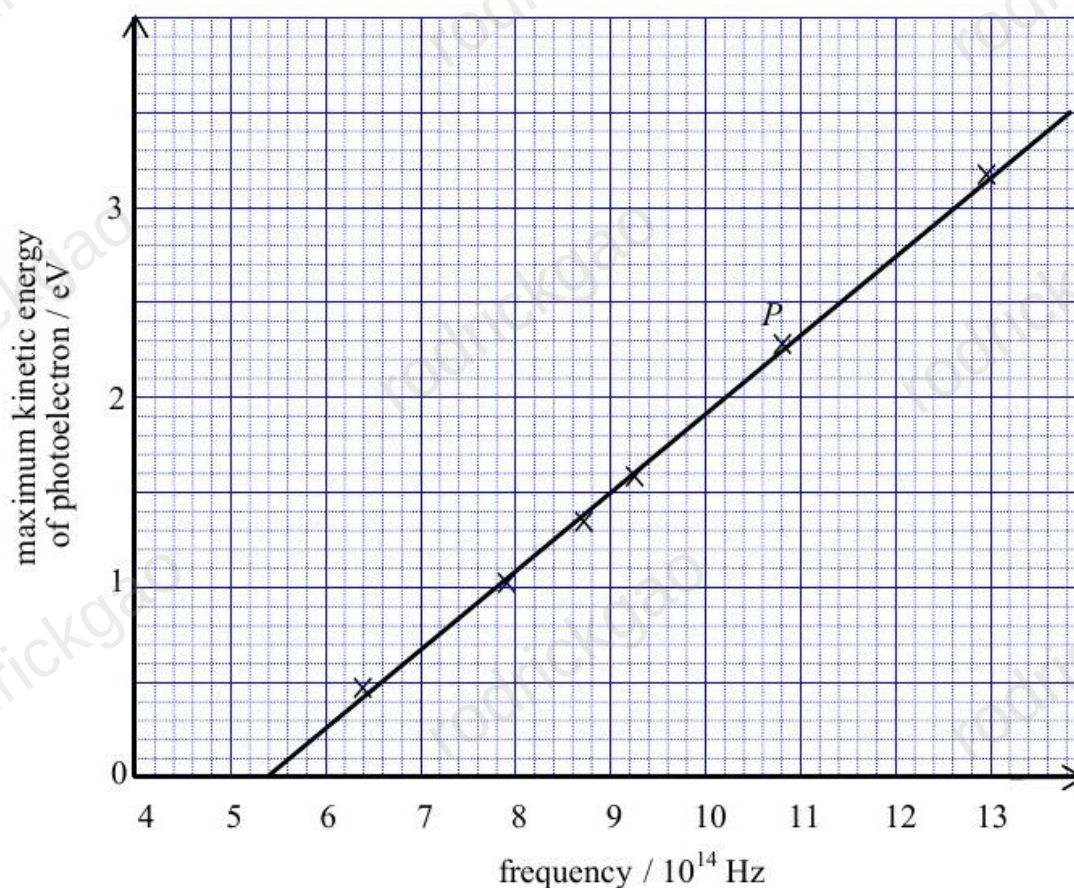


Figure 2.2

- (i) State the part of the electromagnetic spectrum from which radiation was used in obtaining data point P. (1 mark)

- (ii) Find the slope of the graph and deduce its physical meaning. (3 marks)
- (iii) Calculate the work function of sodium in eV. (2 marks)
- (c) State the change, if any, in the graph obtained if the experiment is repeated with electromagnetic radiations of lower intensity. Explain. (2 marks)

Official marking scheme

2. (a)	Increase / adjust the voltage just until there is no current passing through the circuit (no photoelectrons to complete the circuit), record the voltage V_s which gives the maximum kinetic energy of the photoelectrons $= eV_s$	1A 1A	2
(b) (i)	Ultra-violet (UV) radiation (light) (~ 278 nm)	1A	1
			$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10.8 \times 10^{14}} \approx 277.8 \text{ nm}$
(ii)	slope of the graph $= \frac{3.3 - 0}{(13.4 - 5.4) \times 10^{14}}$ $= 4.125 \times 10^{-15} \text{ eV s } (= 6.6 \times 10^{-34} \text{ J s})$	1M 1A	
	The slope gives the Planck constant h .	1A	3
(iii)	Threshold frequency $f_0 = 5.4 \times 10^{14} \text{ Hz}$ Work function of sodium $= hf_0$ $= (6.6 \times 10^{-34}) \times (5.4 \times 10^{14})$ $= 3.564 \times 10^{-19} \text{ J}$ $= 2.2275 \text{ (eV)}$	1M 1A	Accept 2.20 eV to 2.40 eV
	Or $hf_0 = (4.125 \times 10^{-15}) \times (5.4 \times 10^{14})$ $= 2.2275 \text{ (eV)} \approx 2.23 \text{ (eV)}$	1M 1A	2
(c)	Unchanged, i.e. the same graph.	1A	
	The (maximum) kinetic energy depends on the energy of each photon, which is proportional to the frequency of radiation.	1A	
	Or Maximum kinetic energy of the photoelectrons / energy of photon will not be affected.	1A	2

(a) Increase the retarding voltage until the photocurrent just falls to zero. The stopping potential V_s gives $K_{\max} = eV_s$.

(b) (i) Point P corresponds to ultraviolet radiation ($\lambda \approx 278$ nm).

(ii) Slope $= (3.3 - 0) / [(13.4 - 5.4) \times 10^{14}] = 4.125 \times 10^{-15} \text{ eV s} = 6.60 \times 10^{-34} \text{ J s}$, i.e. Planck's constant.

(iii) With $f_0 = 5.4 \times 10^{14} \text{ Hz}$, $\phi = hf_0 = 2.23 \text{ eV}$.

(c) The graph is unchanged: lower intensity reduces the number of emitted electrons/current, but $K_{\max}$ depends on photon frequency, not intensity.

Section C: Energy and Use of Energy

Multiple-choice questions

3.1

Hybrid cars are usually equipped with combustion engines with smaller power than those in petrol cars because

- (1) the combustion engine can be supplemented by the motor to provide the maximum power output of hybrid cars.
- (2) combustion engines in hybrid cars are more energy efficient than those in petrol cars.
- (3) the operation cost per unit power output of combustion engines with smaller power is lower.

- A (1) only
- B (3) only
- C (1) and (2) only
- D (2) and (3) only

Official answer: A

The electric motor can supplement the combustion engine when maximum power is required. The other statements do not necessarily follow, so only (1) explains why the combustion engine can be smaller. Answer:

A.

3.2

Carbon dioxide emission associated with a passenger travelling in a MTR train for 1 km is about 50 g while that for a car is over 200 g. The possible reason(s) is/are

- (1) The MTR train and the car need energy to move the vehicle itself.
- (2) The energy efficiency in transporting a passenger for 1 km is higher in the case of the MTR train.
- (3) No burning of fossil fuel is involved in the production of energy for the MTR train.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

Official answer: B

The MTR uses less energy per passenger-kilometre because many passengers share the vehicle's energy use. Electricity generation can still involve fossil fuel, and statement (1) does not explain the difference. Only (2) is correct. Answer: B.

3.3

An air-conditioner with cooling capacity / electrical power input = 2 has a cooling capacity of 746 W. Estimate the rate at which thermal energy is being released to the environment outside.

A 373 W

B 746 W

C 1119 W

D 1492 W

Official answer: C

$\text{COP} = Q_{\text{cool}}/W_{\text{in}} = 2$, so $W_{\text{in}} = 746/2 = 373$ W. The heat released outside is $Q_{\text{out}} = Q_{\text{cool}} + W_{\text{in}} = 746 + 373 = 1119$ W. Answer: C.

3.4

Which of the following contribute(s) to cooking with a microwave oven?

- (1) The energy of microwave is only absorbed by the food's surface and then transferred to its interior by conduction.
- (2) Water molecules are polar due to non-uniform charge distribution within each molecule.
- (3) The electric field of microwave is continuously changing.

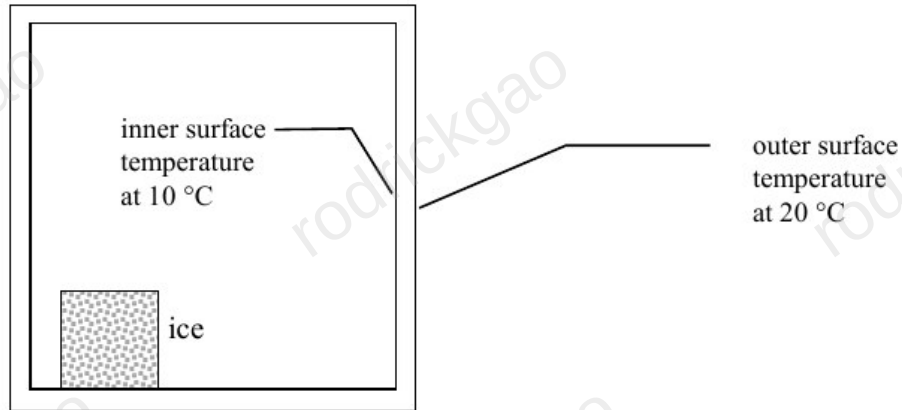
- A (1) only
- B (2) only
- C (1) and (3) only
- D (2) and (3) only

Official answer: D

Water molecules are polar, and the alternating microwave electric field repeatedly turns them, producing heating. Microwaves penetrate the food rather than heating only its surface, so (2) and (3) are correct. Answer: D.

3.5

A piece of ice is placed inside a closed plastic box. The wall of the plastic box is 2 cm thick and the thermal conductivity of plastic is $0.03 \text{ W m}^{-1} \text{ }^{\circ}\text{C}^{-1}$. The temperature of the inner and the outer surfaces of the wall are 10°C and 20°C respectively. What is the rate of heat exchange (per unit area) between the inside and outside of the box and the direction of heat flow?



	Rate of heat exchange	Direction of heat flow
A	15 W m^{-2}	into the box
B	15 W m^{-2}	out of the box
C	36 W m^{-2}	into the box
D	36 W m^{-2}	out of the box

Official answer: A

Heat flux = $k\Delta T/L = 0.03(20-10)/0.02 = 15 \text{ W m}^{-2}$. Heat flows from the warmer outside surface into the box.

Answer: A.

3.6

Which statement below about a house and its features MUST BE correct?

- A Thermal conduction is less efficient in materials that are darker in colour.
- B It is more desirable for the Overall Thermal Transfer Value (OTTV) of a house to be larger in cold areas.
- C Wall painted in white reduces the Overall Thermal Transfer Value (OTTV) of a house.
- D Wall painted in white increases the reflection of sunlight.

Official answer: D

A white wall reflects more incident sunlight. Colour does not determine thermal conductivity, and the other OTTV statements are not necessarily true as worded. Answer: D.

3.7

A wind turbine generator delivers a certain electric power with a wind blowing normal at speed v . If the length of the turbine blades is increased by 25% and the overall efficiency of the turbine generator remains the same, estimate the wind speed blowing normal to the turbine that gives the same electric power.

- A 0.59 v
- B 0.64 v
- C 0.86 v**
- D 0.93 v

Official answer: C

For unchanged efficiency, wind power $P \propto Av^3 \propto L^2v^3$. Thus $v'/v = (1/1.25^2)^{1/3} = 0.862 \approx 0.86$. Answer: C.

3.8

In a nuclear reactor, uranium-235 is used as the fission fuel to generate electricity. When uranium-235 undergoes nuclear fission, 0.08% of its mass is converted to energy. If 20% of this energy becomes electrical energy, estimate the amount of uranium-235 used per second for generating 500 MW of electric power.

- A 1.4×10^{-6} kg
- B 3.5×10^{-5} kg**
- C 8.2×10^{-3} kg
- D 1.0×10^{-2} kg

Official answer: B

Electrical power = $0.20(0.0008)\dot{m}c^2$. Hence $\dot{m} = 500 \times 10^6 / [0.20 \times 0.0008 \times (3.0 \times 10^8)^2] = 3.47 \times 10^{-5}$ kg s⁻¹. Answer: B.

Question 3: Structured question

- (a) (i) Incandescent lamps are far less energy efficient than other light sources like fluorescent lamps or light emitting diodes (LEDs). Explain why this is so in terms of how incandescent lamps produce light. (2 marks)
- (ii) Two light sources of identical size and shape emit white light and green light respectively. If the light output power of them is the same, briefly explain which light source looks brighter. (2 marks)
- (b) Figure 3.1 illustrates the simplified lighting arrangement within a tunnel in which two lamps each of luminous flux 10000 lumens are installed on the ceiling.

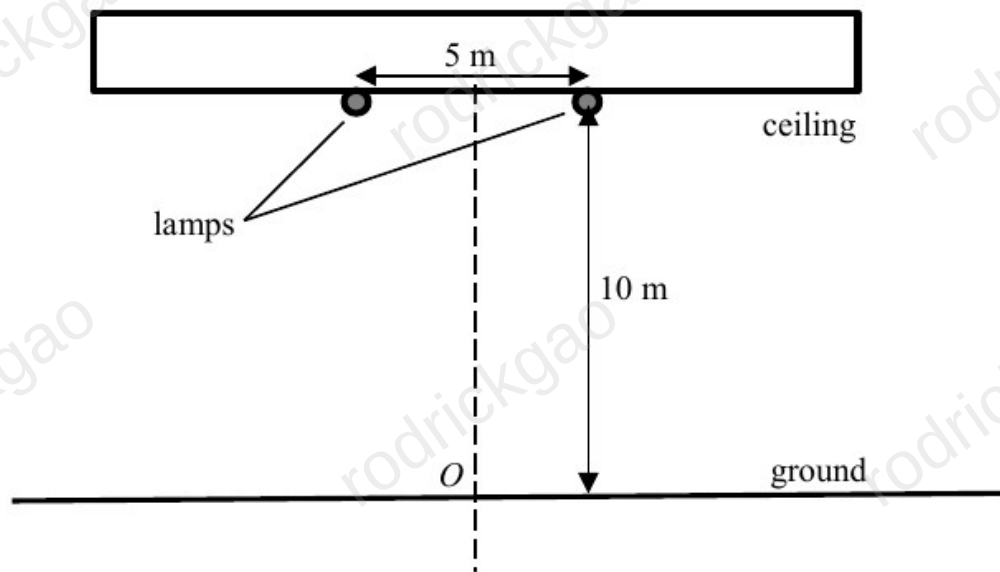
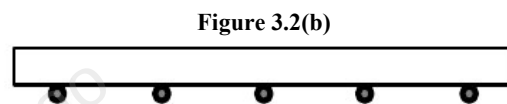
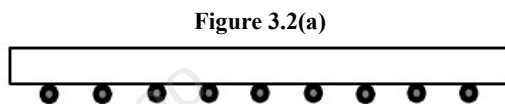


Figure 3.1

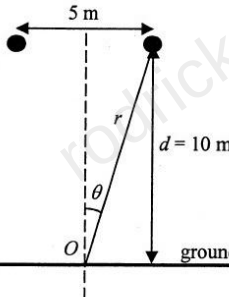
- (i) Calculate the illuminance around point O on the ground mid-way between the two lamps. Take the lamps as point light sources and reflection is assumed negligible. (3 marks)
- (ii) The specification of two kinds of lamps A and B are given below. In terms of efficacy, recommend which one the tunnel company should choose. (1 mark)

Lamp	Rated power	Luminous flux
A	150 W	11000 lumens
B	135 W	10000 lumens

- (iii) Figures 3.2(a) and 3.2(b) illustrate two arrangements of tunnel lights. In Figure 3.2(a), more lamps each with relatively lower luminous flux are used. In Figure 3.2(b), fewer lamps each with relatively higher luminous flux are used. The resultant average illuminance on the ground is the same in both cases. State one advantage and one disadvantage of the arrangement in Figure 3.2(a) over that in Figure 3.2(b). (2 marks)



Official marking scheme

3. (a) (i) Incandescent lamps: by heating a (tungsten) filament to a high temperature / red hot via joule heating (of a current), most of the energy becomes thermal energy / heat or only a small portion is converted to light output.	1A 1A 2	
(ii) As the eye is most sensitive to green light, a green light source (having the same light output power) would appear brighter compared to a white one (comprises of different colours).	1A 1A 2	
(b) (i) Luminous flux of each lamp: $E = 10000$ lumens $\tan \theta = \frac{2.5}{10}$ (or $\cos \theta = \frac{10}{\sqrt{10^2 + 2.5^2}}$) $\theta = 14.036243^\circ \approx 14.0^\circ$ $I = \frac{E \times \cos^3 \theta \times 2}{4\pi d^2}$ Or $\frac{E \times \cos \theta \times 2}{4\pi r^2}$ $= 14.532045$ lux ≈ 14.5 lux or lx	1M 1M 1A 3	
(ii) Efficacy (A) = $\frac{11000}{150} = 73.333333 \text{ lm W}^{-1}$ Efficacy (B) = $\frac{10000}{135} = 74.074074 \text{ lm W}^{-1}$ Lamp B is recommended.	1A 1	
(iii) Advantages: ➤ Variation of illuminance smaller ➤ Effect of individual lamp failure smaller ➤ Less glare Disadvantages: ➤ More frequent change / replacement of lamps ➤ More expensive as installation cost increases ➤ More wiring involved ➤ More installation time	1A Any ONE 1A Any ONE 2	

(a) (i) An incandescent lamp heats a filament to a very high temperature; most input becomes infrared/thermal energy and only a small fraction is visible light.

(ii) For equal radiant power, green appears brighter because the eye is most sensitive near green wavelengths.

(b) (i) For each lamp, horizontal offset = 2.5 m, height = 10 m and $r^2 = 106.25 \text{ m}^2$. Total illuminance = $2[10000/(4\pi r^2)](10/r) = 14.5 \text{ lx}$.

(ii) $\eta_A = 11000/150 = 73.3 \text{ lm W}^{-1}$ and $\eta_B = 10000/135 = 74.1 \text{ lm W}^{-1}$; choose B.

(iii) More lower-flux lamps give more uniform illuminance, less glare and less effect from one failure, but require more wiring, installation and maintenance.

Section D: Medical Physics

Multiple-choice questions

4.1

The refractive indices of air and the components of the eye are listed below. Which pair of media provides the most bending of light due to refraction?

Medium	Refractive index
air	1.00
cornea	1.38
aqueous humour	1.34
surface of lens	1.39
centre of lens	1.41

- A surface of lens – centre of lens
- B aqueous humour – surface of lens
- C cornea – aqueous humour
- D air – cornea**

Official answer: D

The largest refractive-index change is from air (1.00) to cornea (1.38), so that interface bends light most strongly. Answer: D.

4.2

John's vision has a near-point distance of 20 cm and a far-point distance of 250 cm. If he wears spectacles that correct his far point to infinity, what should the type of spectacles be and what is his near-point distance after wearing the spectacles?

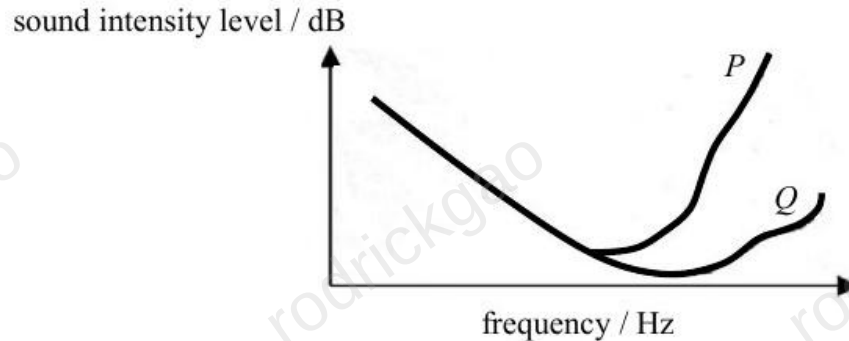
	Type of spectacles	Near-point distance
A	convex lens	18.5 cm
B	concave lens	18.5 cm
C	convex lens	21.7 cm
D	concave lens	21.7 cm

Official answer: D

Correcting a far point of 250 cm requires a concave lens with $f = -250$ cm. For the corrected near point, $1/(-250) = 1/u + 1/(-20)$, giving $u = 21.7$ cm. Answer: D.

4.3

The graph below shows the threshold of hearing for two persons P and Q. Which statements below MUST BE correct?



- (1) P and Q have the same hearing sensitivity for sound of very low frequencies.
- (2) P does not usually hear sound of high frequencies unless the sound wave is large in amplitude.
- (3) Q suffers from a hearing loss.

- A (1) and (2) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: A

The graph shows the same very-low-frequency threshold for P and Q, while P needs a high intensity at high frequency and therefore cannot usually hear weak high-frequency sound. The graph alone does not establish that Q has hearing loss. Answer: A.

4.4

A small loudspeaker emits a sound. The sound intensity level measured at 5 m away from it is 40 dB. The power output of the loudspeaker is then reduced to half and the sound intensity level is measured again at 10 m away. By what percentage has the sound intensity level changed?

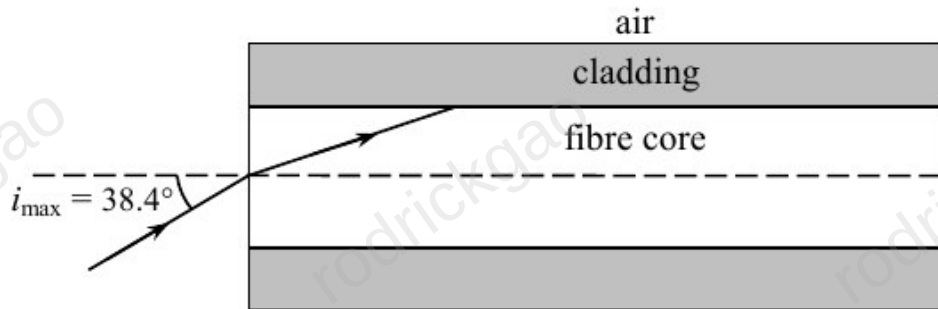
- A 12.5%
- B 15.1%
- C 22.6%
- D 25.0%

Official answer: C

The new intensity is reduced by $(1/2)(5/10)^2 = 1/8$. Thus the level falls by $10\log_{10}8 = 9.03$ dB, from 40.0 to 31.0 dB; the percentage decrease is $9.03/40 = 22.6\%$. Answer: C.

4.5

As shown below the optical fibre used for an endoscope is clad with material of refractive index different from that of the fibre core to avoid light leakage. The maximum angle of incidence $i_{\max}$ is 38.4° for no leakage of light. Find the refractive index of the cladding material. Given: refractive index of the fibre core = 1.55.



- A 1.69
- B 1.42**
- C 1.33
- D 1.16

Official answer: B

For a step-index fibre in air, $NA = \sin i_{\max} = \sqrt{(n_{\text{core}}^2 - n_{\text{clad}}^2)}$. Hence $n_{\text{clad}} = \sqrt{[1.55^2 - \sin^2(38.4^\circ)]} = 1.42$.

Answer: B.

4.6

Which statements about radionuclide imaging (RNI) are correct?

- (1) The functions of organs in a body can be assessed using RNI.
- (2) Thyroid problems can be identified using RNI.
- (3) The spatial resolution of RNI is as good as that of a computed tomography (CT) scan.

- A (1) and (2) only**
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: A

RNI can assess organ function and diagnose thyroid problems, but its spatial resolution is poorer than CT. Thus (1) and (2) are correct. Answer: A.

4.7

A gamma source produces a corrected count rate of 3500 s^{-1} at a point 1 m away. What minimum thickness of concrete is needed to reduce the corrected count rate to less than 200 s^{-1} at the same point? Given: the half-value thickness of concrete is 60.5 mm.

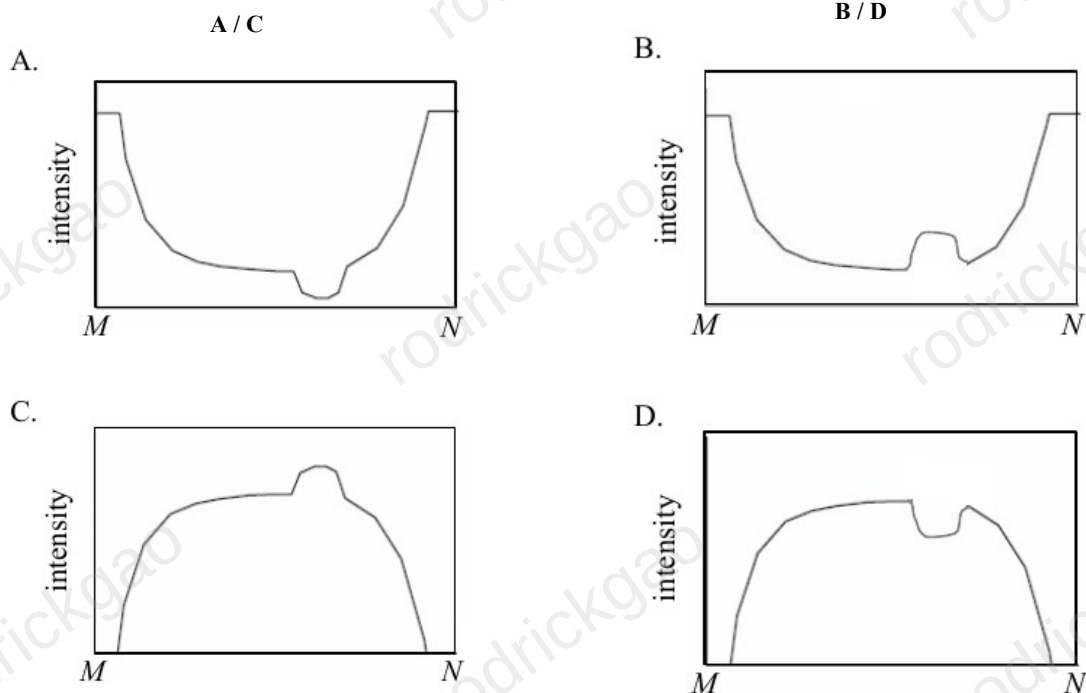
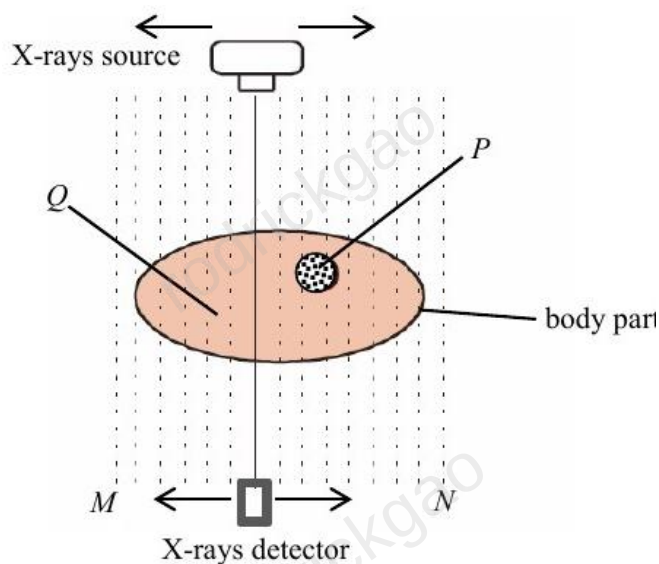
- A 40 cm
- B 25 cm**
- C 15 cm
- D 10 cm

Official answer: B

Required attenuation is $3500/200 = 17.5$. Thickness $x = 60.5 \log_2(17.5) = 250 \text{ mm}$, so a little over 25 cm is required; the matching choice is 25 cm. Answer: B.

4.8

An X-ray source paired with a detector can move horizontally to scan a body part (consisting of tissues P and Q) as shown. Which graph below correctly shows the variation of intensity of the X-rays detected? Given: attenuation coefficient of P is greater than that of Q.



Official answer: A

X-ray intensity follows $I = I_0 e^{(-\mu x)}$. Regions with more material or with tissue P (larger μ) give lower detected intensity; matching the path lengths across the scan gives graph A. Answer: A.

Question 4: Structured question

- (a) Piezoelectric transducers can generate as well as detect ultrasound. Figure 4.1 shows the basic construction of a piezoelectric transducer in which the thickness of the piezoelectric crystal is t .

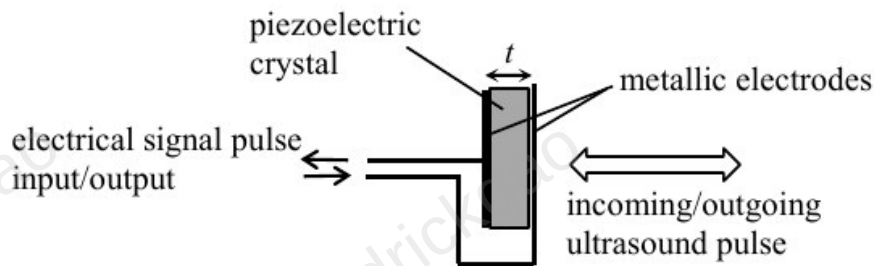
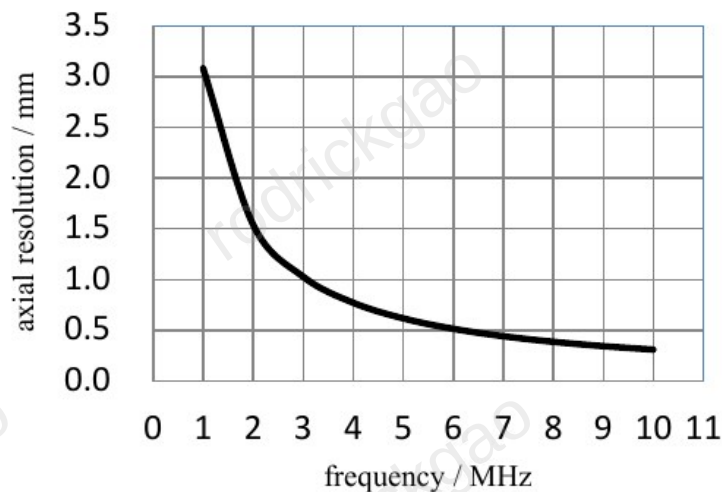


Figure 4.1

- (i) The crystal has a maximum response for a certain frequency f of the electrical signal applied such that t is about half of the wavelength of the ultrasound in the crystal. The speed of ultrasound in the crystal is 4000 m s^{-1} . Estimate f if $t = 0.4 \text{ mm}$. (1 mark)
- (ii) An ultrasound pulse produced by the piezoelectric transducer is reflected back from an interface as an echo. Explain how the transducer detects this echo. (2 marks)
- (b) For medical imaging using an ultrasound beam, the resolution (in mm) along the beam direction (axial direction) in soft tissue varies with the frequency (in MHz) of the ultrasound used as shown below.



Axial resolution versus frequency

- (i) Explain the importance of axial resolution in ultrasound imaging. (2 marks)
- (ii) In fact the axial resolution is inversely proportional to the frequency of ultrasound used. Make use of one or two points of the graph to deduce the axial resolution of imaging when using 12 MHz ultrasound. (2 marks)

- (c) Figure 4.2 shows the set-up for studying a tissue interface underneath a 2 cm thick fat layer and an 8 cm thick soft tissue layer. The speed of ultrasound in the fat layer is 1450 m s^{-1} while that in the soft tissue layer is 1540 m s^{-1} .

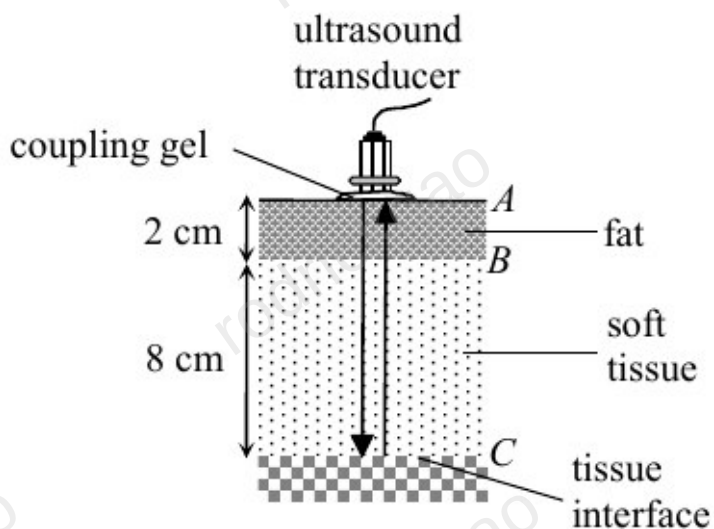


Figure 4.2

- Calculate the echo time T for the ultrasound pulse going to and from the tissue interface at C. (2 marks)
- The depth of the tissue interface is computed using a device which takes the whole echo time T as being spent on propagating to and fro in soft tissue only (i.e. calibrated at a speed of 1540 m s^{-1}). Determine by how much this computed depth differs from the actual depth of the tissue interface. (1 mark)

Official marking scheme

4. (a) (i)	The frequency f is $f = \frac{c}{\lambda} = \frac{4000 \text{ ms}^{-1}}{2 \times 0.4 \times 10^{-3} \text{ m}} = 5 \text{ MHz}$	1A	1	Or resolution = $\frac{k}{\text{frequency}}$ resolution $\times$ frequency = k $\therefore k \approx 3$
(ii)	The piezoelectric crystal is forced to vibrate / change shape when the ultrasound echoes hit the crystal. When the piezoelectric crystal is compressed or extended, it produces a potential difference / voltage across its ends.	1A	1A	
(b) (i)	With better resolution, the ability to detect, and separate, two objects / points at different distances along the path of the beam (axial direction) is higher.	1A	2	
	Or Points at a closer distance can be distinguished as two separate points. Resolution is a measure of the smallest detail (structure) that can be detected in an image, improvement in resolution gives an image with more details (or finer).	1A	1A	
(ii)	Given the relationship (inversely proportional): axial resolution $\times$ frequency = constant i.e. $1.5 \text{ mm} \times 2 \text{ MHz} = 3 \text{ mm MHz}$ (using one data point)	1M	2	
	Thus, at 12 MHz the axial resolution = $\frac{3}{12} = 0.25 \text{ mm}$	1A	2	
(c) (i)	The round-trip travel time of the ultrasound pulse through 2 cm of fat and 8 cm of soft tissue are respectively,	1M		
	$t_1 = \frac{2 \times 2 \text{ cm}}{1.45 \times 10^5 \text{ cm s}^{-1}} = 27.5862069 \text{ } \mu\text{s} \approx 27.6 \text{ } \mu\text{s}$			
	$t_2 = \frac{2 \times 8 \text{ cm}}{1.54 \times 10^5 \text{ cm s}^{-1}} = 103.896104 \text{ } \mu\text{s} \approx 103.9 \text{ } \mu\text{s}$			Accept 131 μs to 132 μs
	Echo time $T = (27.59 + 103.90) \text{ } \mu\text{s}$ $= 131.482311 \text{ } \mu\text{s} \approx 131 \text{ } \mu\text{s}$	1A	2	
(ii)	The computed depth is	1M/1A		
	$\frac{1.54 \times 10^5 \text{ cm s}^{-1} \times 131.48 \text{ } \mu\text{s}}{2} = 10.124138 \text{ cm} \approx 10.12 \text{ cm}$ i.e. the depth of tissue interface is computed 1.2 mm deeper than its actual position.		1	

(a) (i) Since $t = \lambda/2$, $f = 4000/[2(0.4 \times 10^{-3})] = 5.0 \text{ MHz}$.

(ii) The echo strains the crystal, whose piezoelectric response produces a voltage.

(b) (i) Axial resolution is the minimum separable distance along the beam; a smaller value gives finer detail.

(ii) From the graph, resolution $\times$ frequency $\approx 1.5 \times 2 = 3 \text{ mm MHz}$, so at 12 MHz the resolution is $3/12 = 0.25 \text{ mm}$.

(c) (i) $T = 2(0.02/1450 + 0.08/1540) = 1.315 \times 10^{-4} \text{ s} = 131.5 \text{ } \mu\text{s}$.

(ii) The calibrated depth is $1540T/2 = 10.124 \text{ cm}$, about 1.2 mm deeper than the actual 10.0 cm.